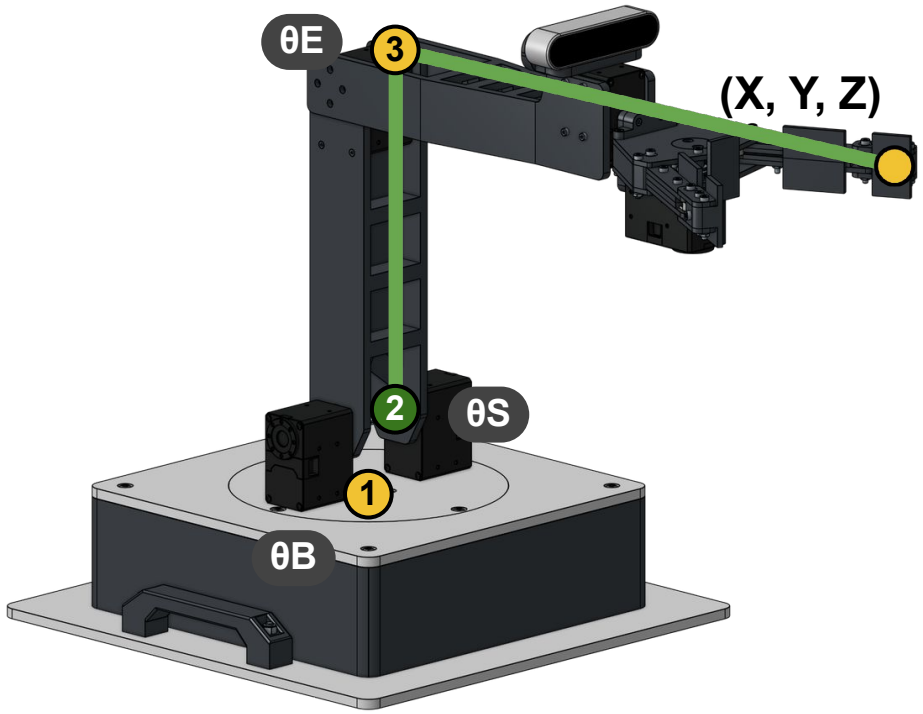
ROBOTIC ARM COURSE · LESSON 4 OF 12

The Denavit-Hartenberg Approach to Forward Kinematics

Turning joint angles into the end effector's position, one matrix per joint

Day 1 · Rotation and movement matrices

Day 2 · Extending to 3D



RECAP · LESSON 3

Where Lesson 3 Left Off

GIVEN

$L_1 = L_2 = 250$ mm

$\theta S = 45°, \theta E = 45°$

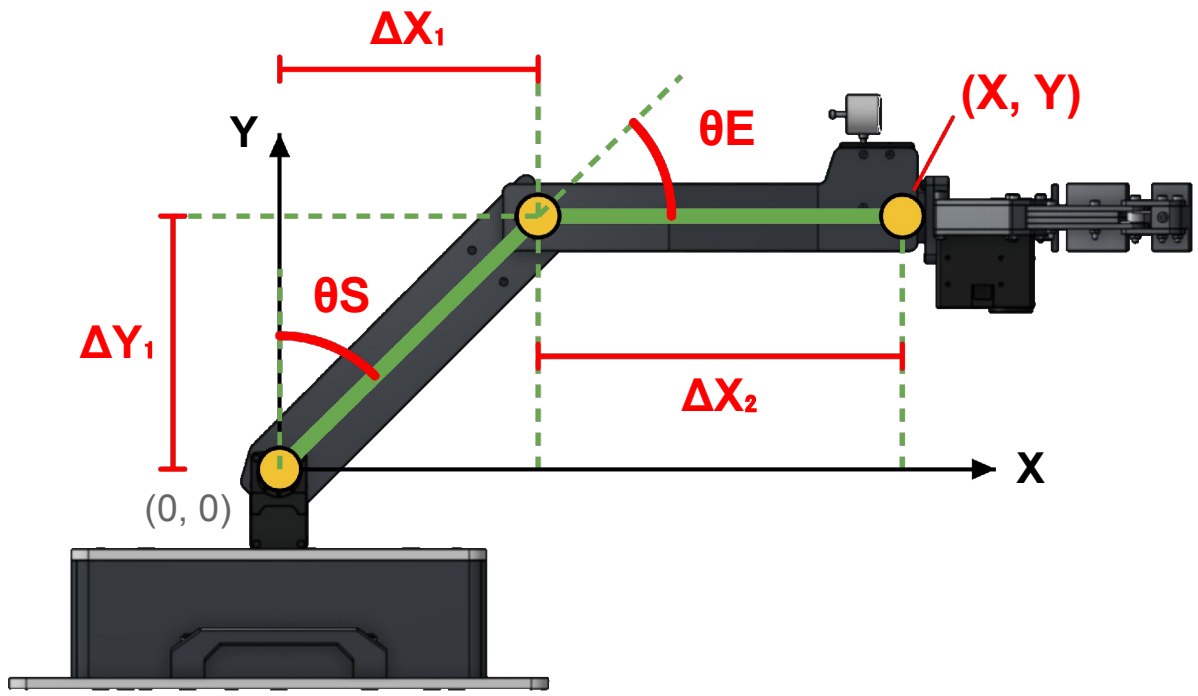
THE TRIANGLES GAVE

$\Delta X_1 = \Delta Y_1 = 176.78$ mm

$\Delta X_2 = 250$ mm, $\Delta Y_2 = 0$ mm

$(X, Y) = (426.78, 176.78)$

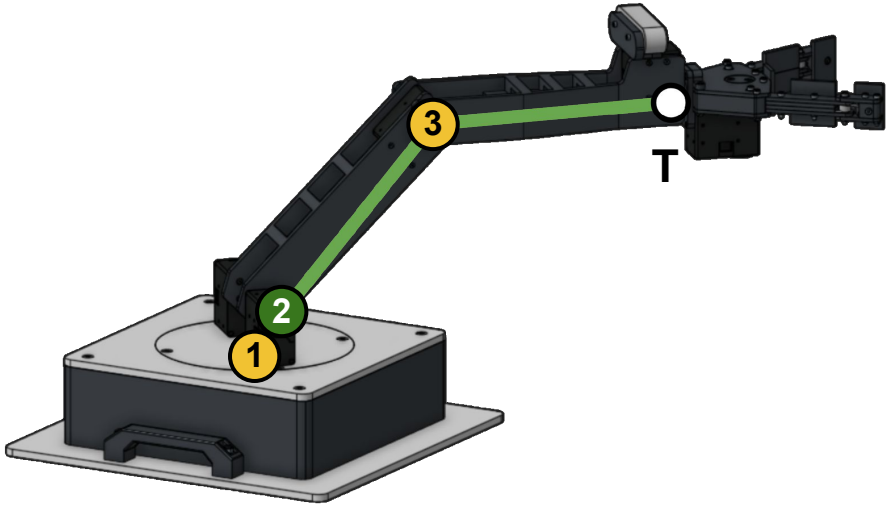
Lesson 3 called the angles θ_1 and θ_2. From now on: **θS** for the shoulder, **θE** for the elbow and **θB** for the base.



WHY MATRICES

One Recipe for Every Joint

Lesson 3 used triangles. Every new joint needed new triangles, and the turning base made the problem 3D.



1 · BASE

A_1

one matrix

×

2 · SHOULDER

A_2

one matrix

×

3 · ELBOW

A_3

one matrix

=

RESULT

T

gives the position

DAY 1

The flat (2D) arm, using rotation and move matrices

DAY 2

The real (3D) arm, using the Denavit-Hartenberg recipe

DAY 1 · ROTATION

Rotating a Point

Spin the point **P = (250, 0)** by 45° around the origin.

Where does it land?

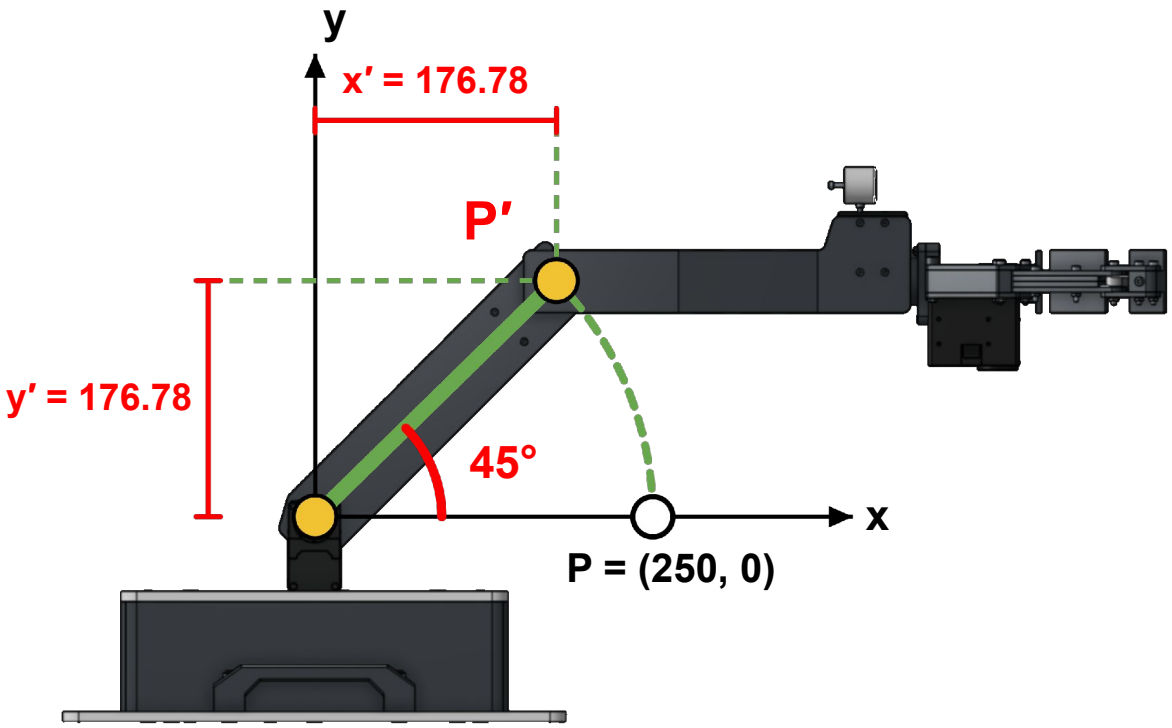
$$x' = 250 \cdot \cos 45° = 176.78$$

$$y' = 250 \cdot \sin 45° = 176.78$$

P′ = (176.78, 176.78)

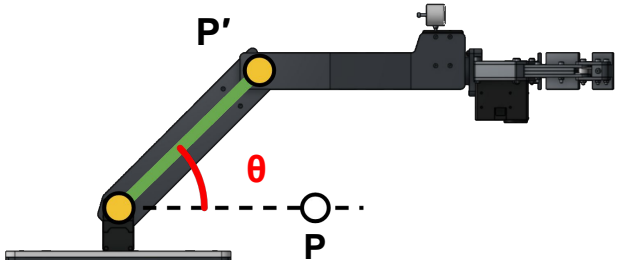
Same numbers as ΔX_1 and ΔY_1 in Lesson 3.

The first link is just a point that was rotated.



DAY 1 · ROTATION

The Rotation Matrix



The two formulas for rotating a point (x, y) by θ fit inside one matrix.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

rotation matrix R(θ) | the point P | the rotated point P′ = (x′, y′)

HOW TO MULTIPLY

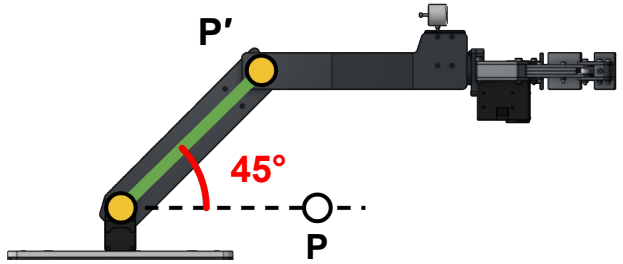
Take one row of the matrix. Multiply it entry by entry with the point, then add. **Row 1** gives x′. **Row 2** gives y′.

WHAT R(θ) DOES

It spins any point by θ, counter-clockwise, around the origin.

DAY 1 · ROTATION

Worked Example: Rotation



Rotate P = (250, 0) by 45°. At 45°, cos and sin are both 0.7071.

$$\begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix} \times \begin{bmatrix} 250 \\ 0 \end{bmatrix} = \begin{bmatrix} 176.78 \\ 176.78 \end{bmatrix}$$

R(45°) P P′

ROW 1 GIVES x′

0.7071 × 250 + (−0.7071) × 0 = 176.78

ROW 2 GIVES y′

0.7071 × 250 + 0.7071 × 0 = 176.78

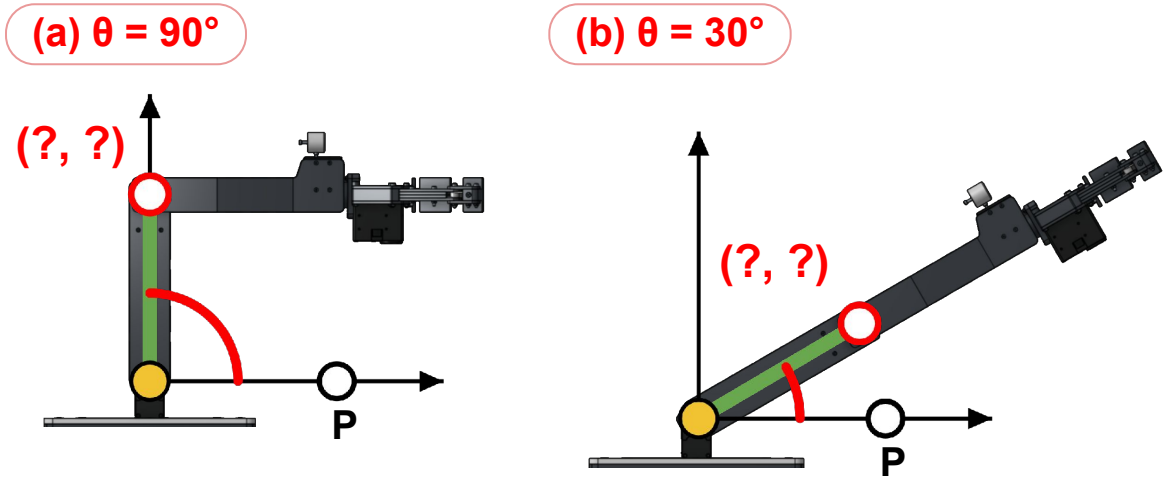
P′ = (176.78, 176.78), the same answer as the cos and sin formulas.

DAY 1 · TRY IT YOURSELF

Try It Yourself #1

Rotate **P = (250, 0)** by each angle. Find P′ = (x′, y′).

(a) θ = 90° **(b) θ = 30°**



SOLUTION

(a) 90°: cos 90° = 0, sin 90° = 1

P′ = (0, 250)

(b) 30°: cos 30° = 0.8660, sin 30° = 0.5

P′ = (216.51, 125.00)

DAY 1 · MOVING A POINT

Moving a Point

Slide the point **P = (250, 0)** to the right by 100.

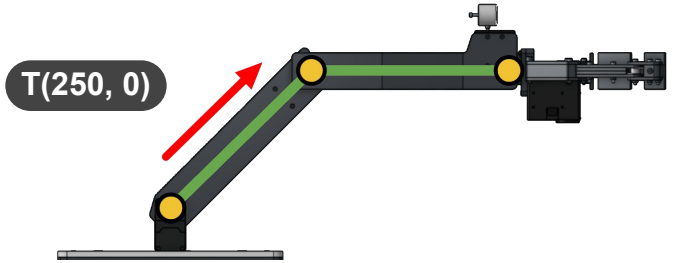
$$x' = x + 100 = 350$$

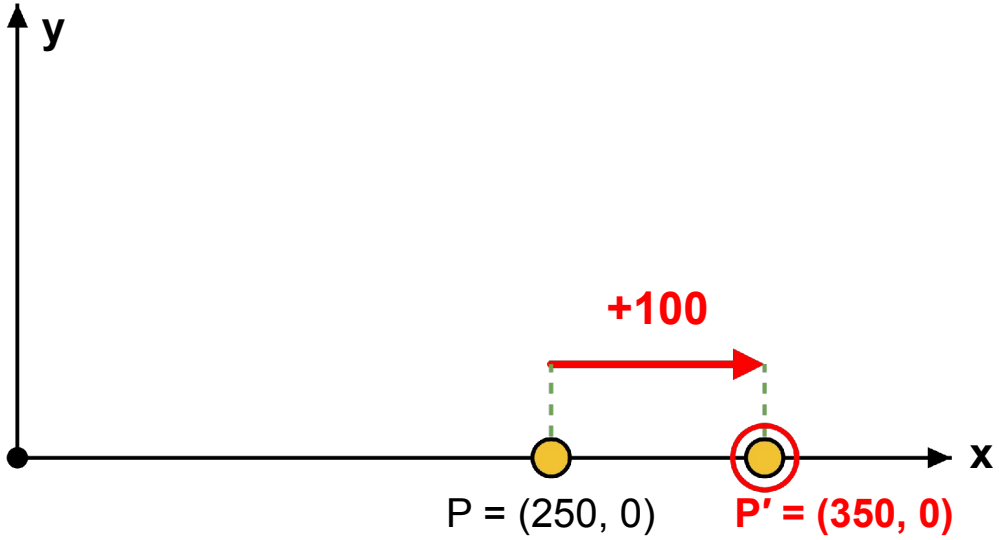
$$y' = y + 0 = 0$$

P′ = (350, 0)

THE PROBLEM

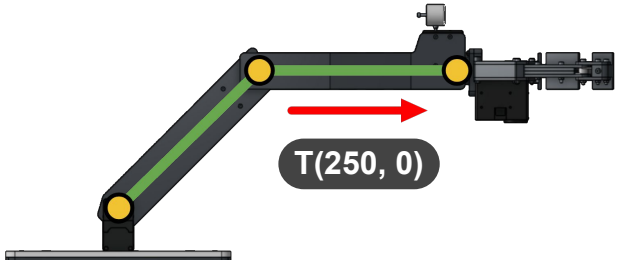
Rotating is multiplying. Moving is adding. Our 2×2 matrix can only multiply, so it cannot move a point.





DAY 1 · MOVING A POINT

The Extra 1



Write every point with an extra 1: (x, y) becomes (x, y, 1). Now moving is a multiplication too, using a 3×3 matrix.

$$\begin{bmatrix} 1 & 0 & 100 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 250 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 350 \\ 0 \\ 1 \end{bmatrix}$$

T(100, 0) P P′

Row 1: 1 · 250 + 0 · 0 + 100 · **1** = 350. The extra 1 is what picks up the 100.

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

R(θ)

spin by θ

$$\begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix}$$

T(dx, dy)

move dx right, dy up

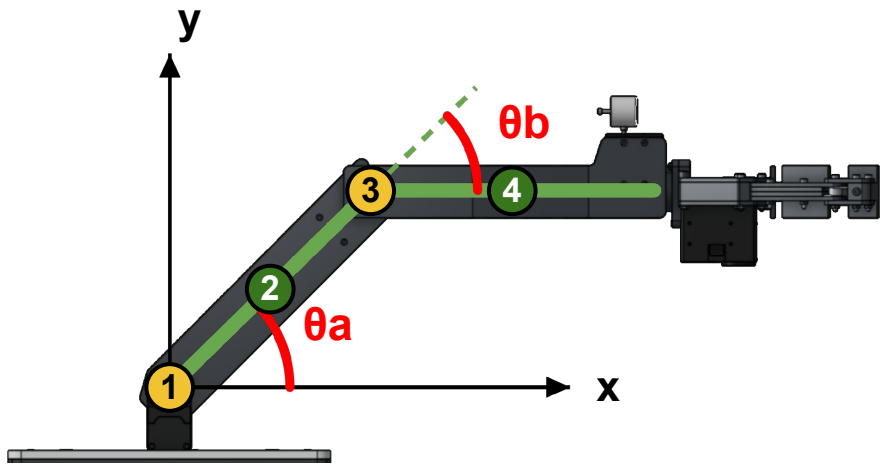
DAY 1 · THE 2D ARM

The Arm as a Chain of Matrices

1. Turn at the shoulder — **R(θa)**
2. Travel along the upper arm — **T(250, 0)**
3. Turn at the elbow — **R(θb)**
4. Travel along the forearm — **T(250, 0)**

M = R(θa) · T(250, 0) · R(θb) · T(250, 0)

Read left to right, it is the story of the arm.



DAY 1 · THE 2D ARM

Converting the Arm's Angles

SHOULDER

$\theta a = 90° - \theta S$

A matrix counts angles counter-clockwise from the x-axis. θS counts clockwise from the vertical.

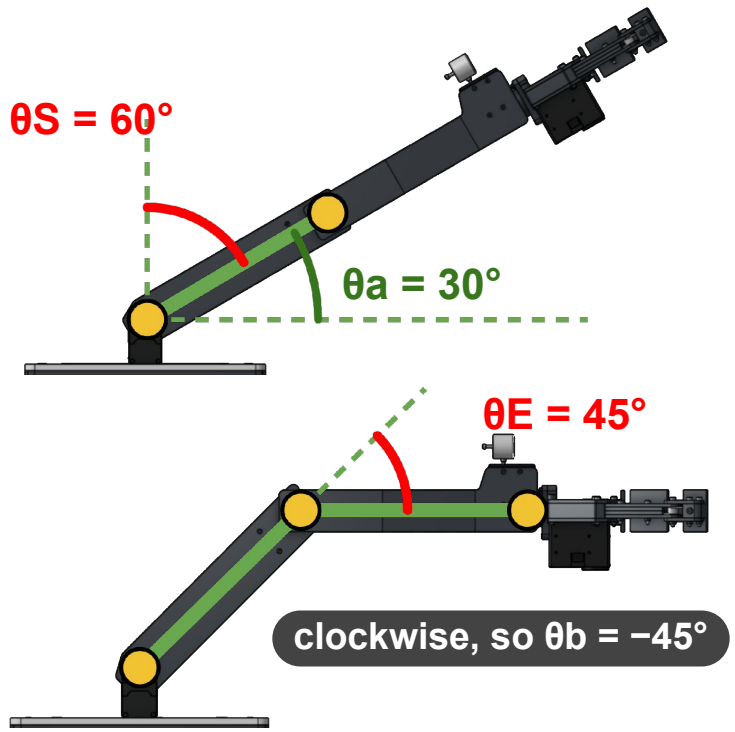
ELBOW

$\theta b = -\theta E$

The forearm bends clockwise. A matrix counts counter-clockwise, so the sign flips.

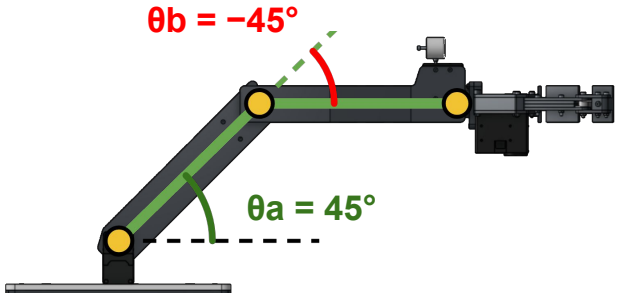
θS = 60° gives θa = 30°

θE = 45° gives θb = −45°



DAY 1 · THE 2D ARM

Worked Example: Setup



Find the end effector for **θS = 45°** and **θE = 45°**, with **$L_1 = L_2 = 250$ mm**.

CONVERT THE ANGLES

θa = 90° − 45° = 45°

θb = −45°

THE MOVES

Both links are 250 mm

Both moves are T(250, 0)

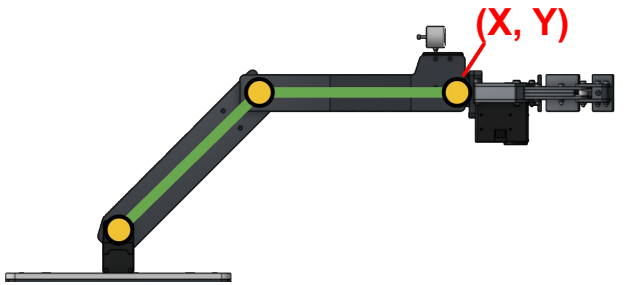
THE CHAIN, APPLIED TO THE POINT (0, 0, 1)

R(45°) ·	T(250, 0) ·	R(−45°) ·	T(250, 0) ·	(0, 0, 1)
step 4	step 3	step 2	step 1	start

The story reads left to right; the arithmetic starts at the right, with the matrix next to the point.

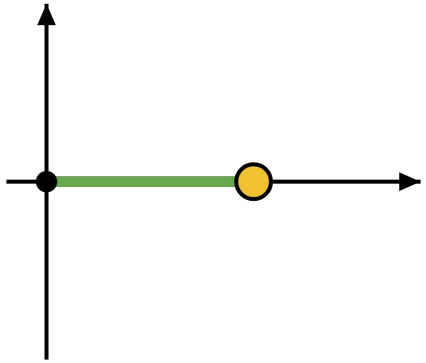
DAY 1 · THE 2D ARM

Worked Example: Right to Left



Each step is one matrix times the current point.

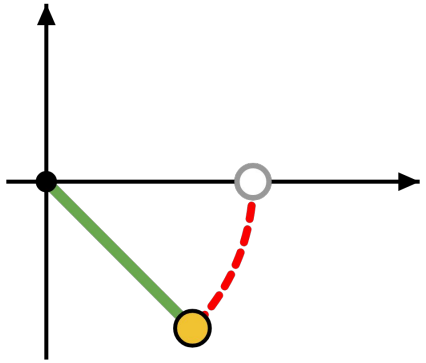
STEP 1 · T(250, 0)



Lay the forearm out along x

(250, 0)

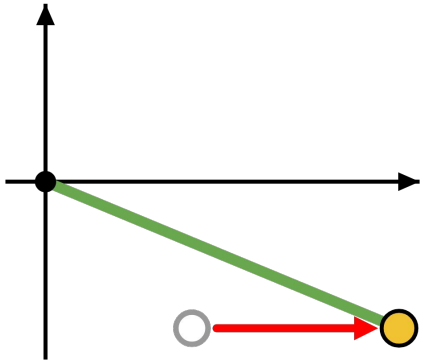
STEP 2 · R(−45°)



Bend the elbow

(176.78, −176.78)

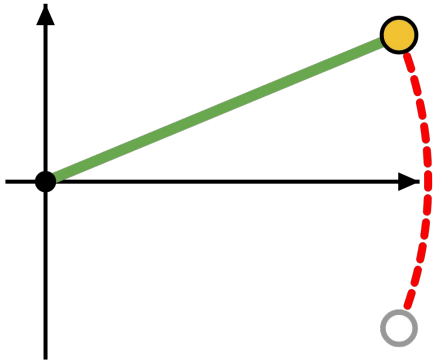
STEP 3 · T(250, 0)



Add the upper arm

(426.78, −176.78)

STEP 4 · R(45°)



Swing everything up to the shoulder angle

(426.78, 176.78)

DAY 1 · THE 2D ARM

Completed Variables

$L_1 = 250$ mm

$L_2 = 250$ mm

$\theta S = 45°$

$\theta E = 45°$

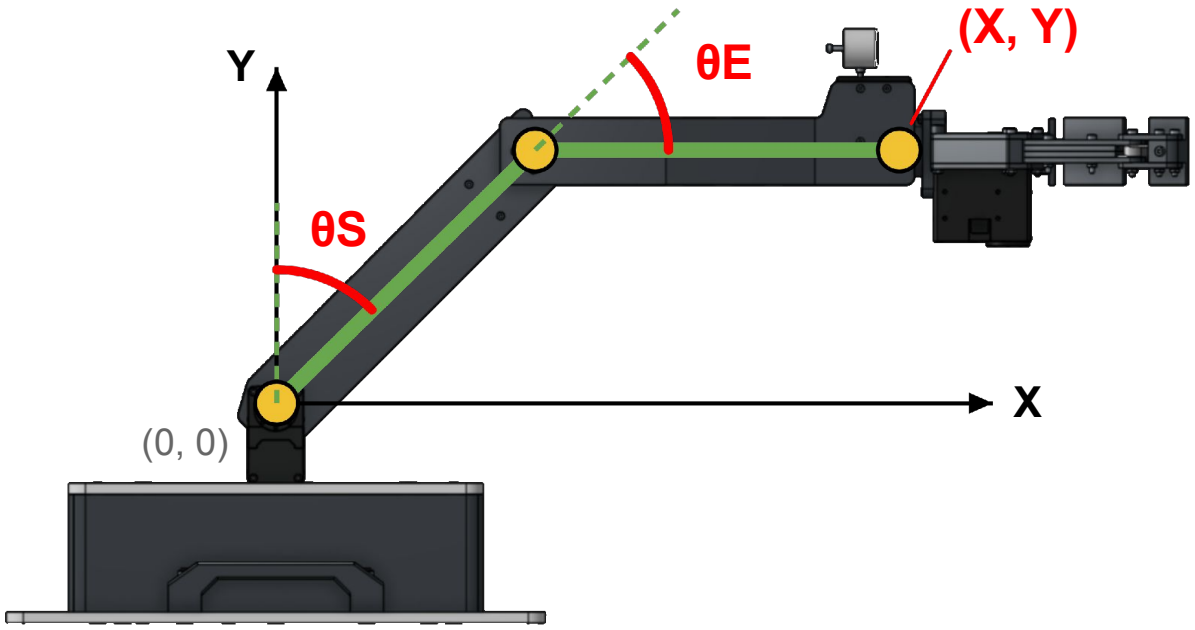
$\theta a = 45°$

$\theta b = -45°$

$(X, Y) = (426.78, 176.78)$

Lesson 3's triangles gave (426.777, 176.777).

The matrices agree.



DAY 1 · TRY IT YOURSELF

Try It Yourself #2

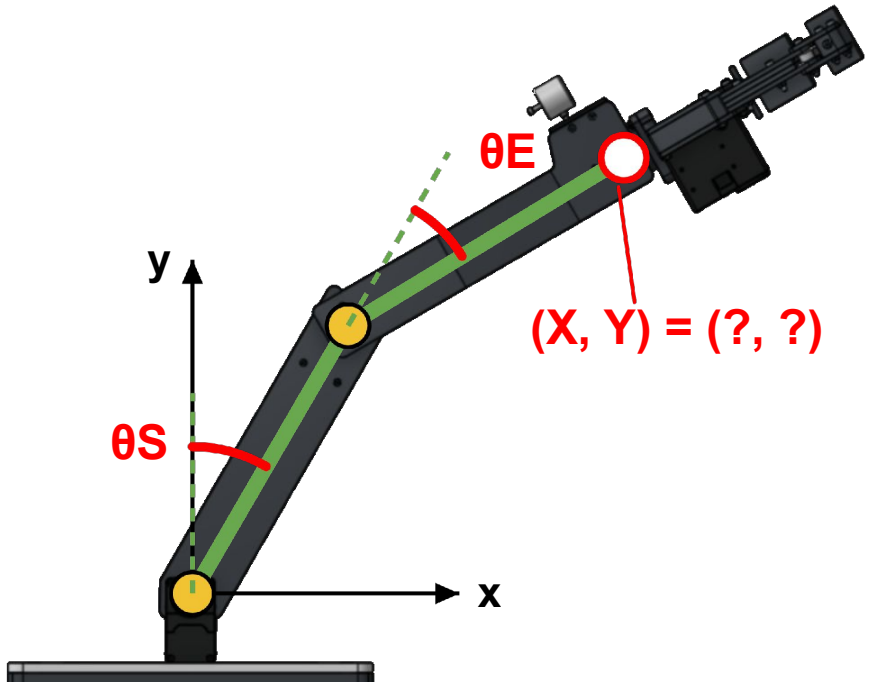
Find the end effector (X, Y).

L_1 = 250 mm

L_2 = 250 mm

θS = 30°

θE = 30°



SOLUTION

θa = 90° − 30° = 60°, θb = −30°

(X, Y) = (341.51, 341.51)

The same answer as Try It Yourself in Lesson 3.

DAY 1 · TRY IT YOURSELF

Try It Yourself #3

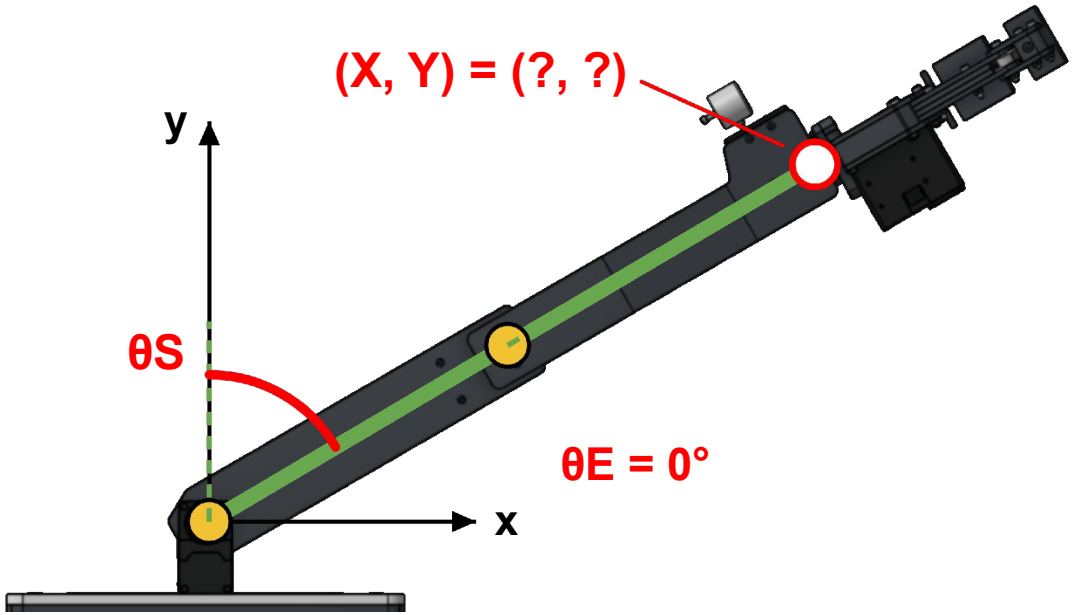
Find the end effector (X, Y).

L_1 = 250 mm

L_2 = 250 mm

θS = 60°

θE = 0°



SOLUTION

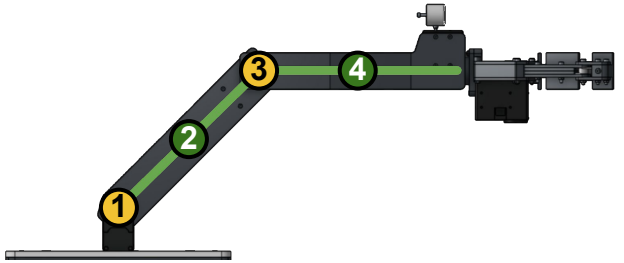
θa = 90° − 60° = 30°, θb = 0°

(X, Y) = (433.01, 250.00)

A 0° turn changes nothing, so the arm is one straight line.

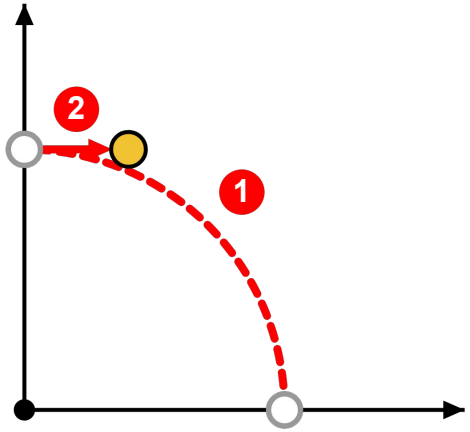
DAY 1 · ORDER MATTERS

Order Matters



A · ROTATE, THEN MOVE

T(100, 0) · R(90°) · P



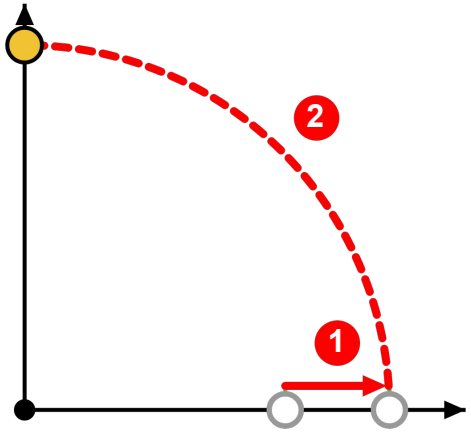
1 · Rotate: (0, 250)

2 · Move: (100, 250)

(100, 250)

B · MOVE, THEN ROTATE

R(90°) · T(100, 0) · P



1 · Move: (350, 0)

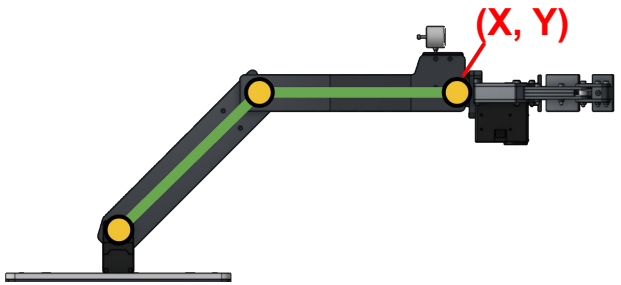
2 · Rotate: (0, 350)

(0, 350)

Same two matrices, different order, different answer. In the arm's chain, the order of the matrices must match the order of the joints.

DAY 1 · RECAP

Day 1 Recap



ROTATE

R(θ)

Spins a point by θ around the origin.

MOVE

T(dx, dy)

Moves a point. The extra 1 makes it a matrix.

CHAIN

M = R · T · R · T

One matrix per step. Multiply from the right.

ORDER

R · T is not T · R

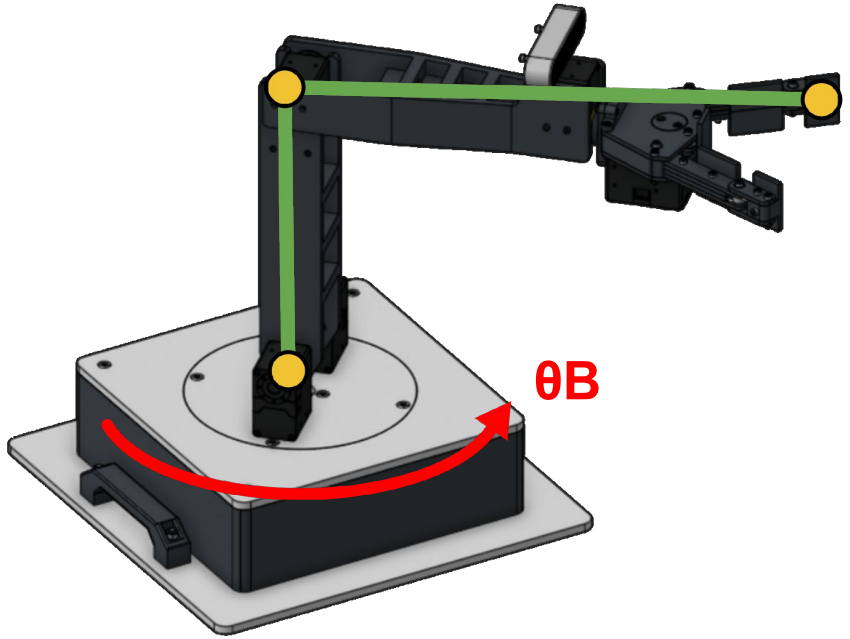
Swap two matrices and the answer changes.

Next time: the real arm turns at the base too, so it needs 3D.

DAY 2

Extending to 3D

Day 1 gave us a recipe for a flat arm. Now the base turns, and the same recipe has to work in three dimensions.



DAY 1 IN ONE LINE

M = R(θa) · T(250, 0) · R(θb) · T(250, 0) → (426.78, 176.78)

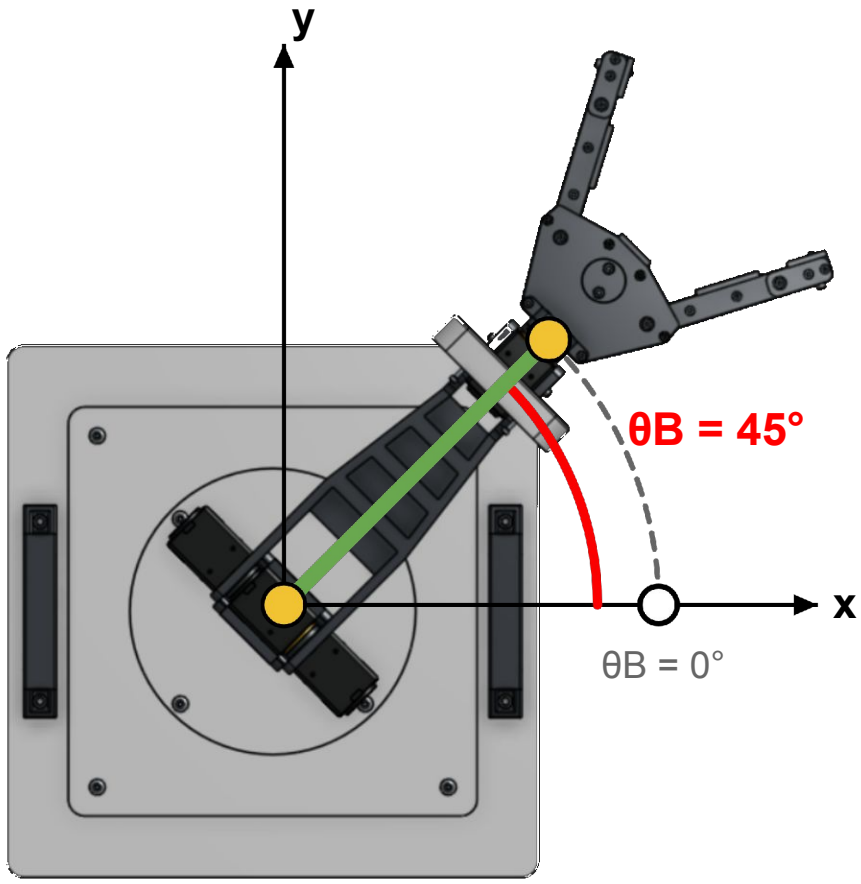
DAY 2 · THE BASE

The Base Turns the Whole Arm

1. Day 1's flat arm sits in one vertical plane.
2. The base spins that whole plane around the vertical axis, by θB.
3. So we need 3D: a point is (x, y, z), and each matrix is 4 × 4.

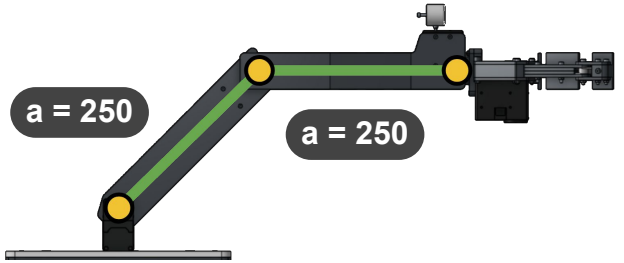
ONE CHANGE OF HABIT

From now on, z points up. Lesson 3 called the up direction Y.



DAY 2 · THE FOUR NUMBERS

Four Numbers Describe Each Joint



Each joint is four simple moves, and each move has one number. The values below are for joints 1, 2, 3: base, shoulder, elbow.

TURN AROUND z

θ

joint angle

How far the joint has turned.

OUR ARM

the only number that changes

SLIDE ALONG z

d

offset

How far the next link is shifted along the joint's axis.

OUR ARM

0 · 0 · 0

SLIDE ALONG x

a

link length

How long the link is.

OUR ARM

0 · 250 · 250 mm

TURN AROUND x

α

link twist

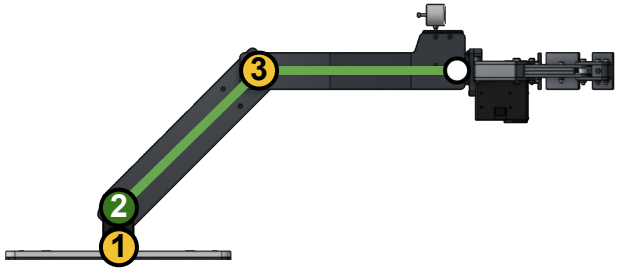
How much the next joint's axis is tilted.

OUR ARM

90° · 0° · 0°

DAY 2 · THE DH MATRIX

One Matrix per Joint



$A = Rz(\theta) \cdot Tz(d) \cdot Tx(a) \cdot Rx(\alpha)$

turn about z, slide along z, slide along x, turn about x

$$A = \begin{bmatrix} \cos\theta & -\sin\theta\cos\alpha & \sin\theta\sin\alpha & a\cos\theta \\ \sin\theta & \cos\theta\cos\alpha & -\cos\theta\sin\alpha & a\sin\theta \\ 0 & \sin\alpha & \cos\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Top-left 3 × 3
how the new frame is turned

Last column
where the new frame's origin is

Bottom row
always 0 0 0 1

We never expand this by hand. We plug in one row of the DH table.

DAY 2 · FRAMES

Placing a Frame on Each Joint

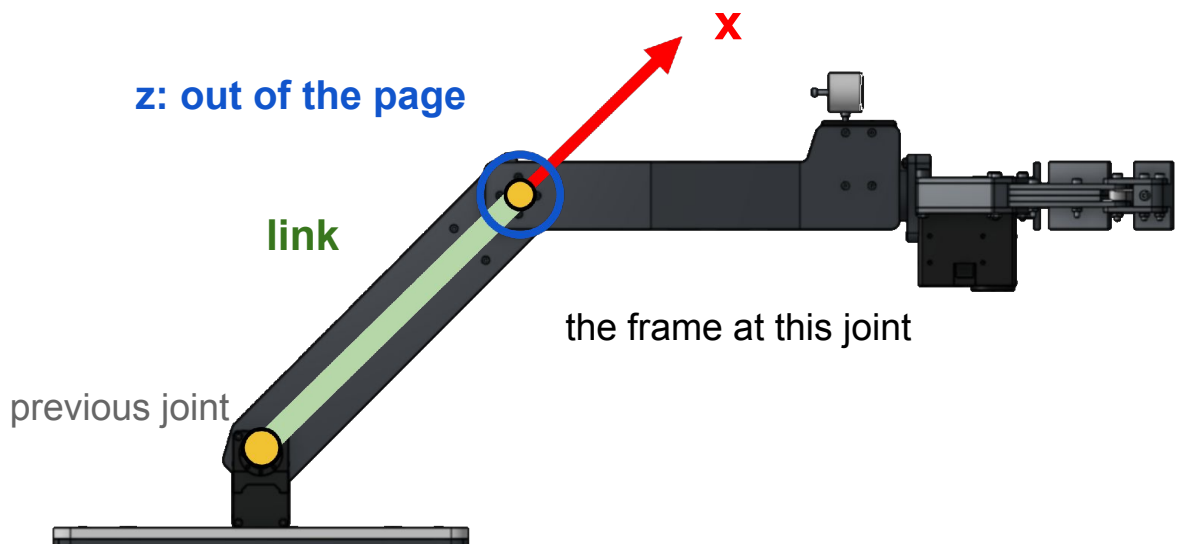
z z points along the axis the joint turns around.

x x points along the link, away from the previous joint.

y is never needed. Once x and z are set, y follows.

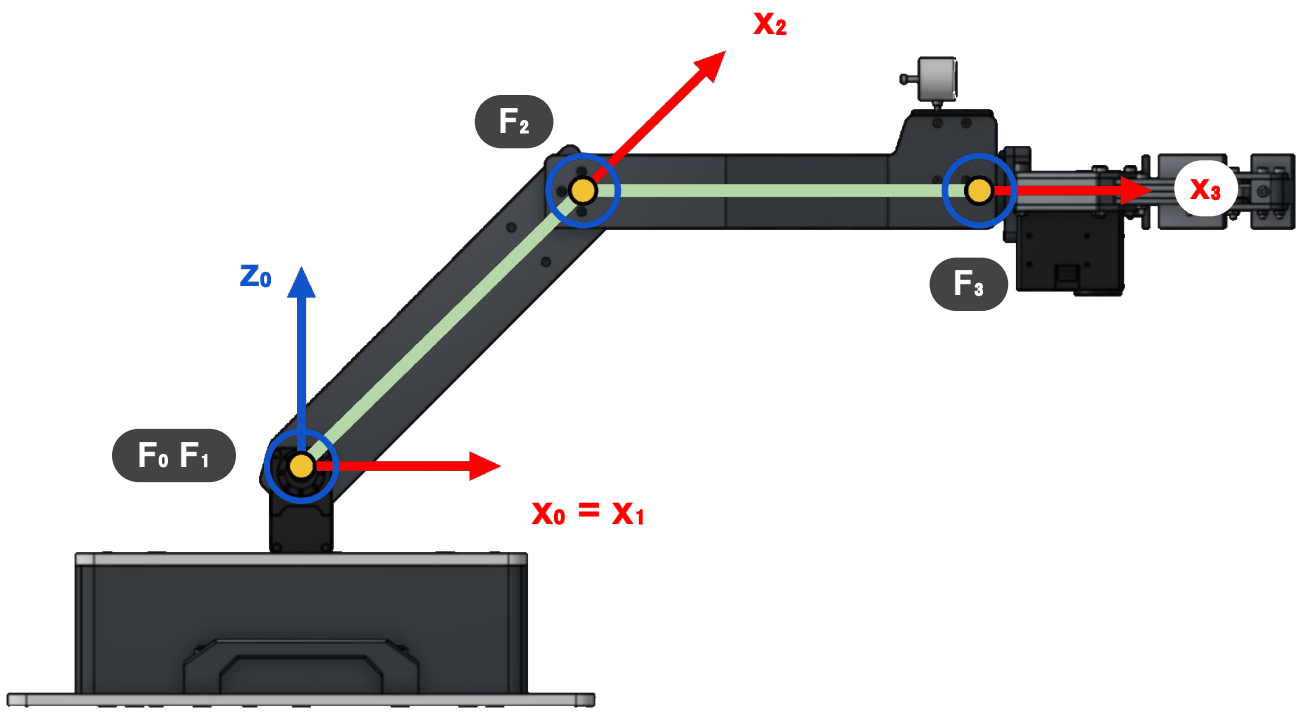
NUMBERING

Frame 0 is fixed on the base. Each later frame sits at the end of a link.



DAY 2 · FRAMES

Frames on Our Arm



F0 F1 Base and shoulder

z0 points up (the base axis). z1 comes out of the page (the shoulder axis). Both frames share one origin, since a = 0 and d = 0 for the base.

F2 Elbow

z2 comes out of the page (the elbow axis). x2 runs along the upper arm.

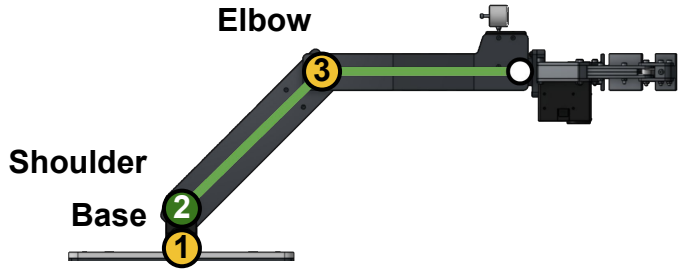
F3 Tip

z3 copies z2. x3 runs along the forearm.

DAY 2 · THE DH TABLE

Fill In the DH Table

Joint	θ	d	a	α
1 · Base	?	?	?	?
2 · Shoulder	?	?	?	?
3 · Elbow	?	?	?	?



ROW i

Row i takes you from frame i − 1 to frame i.

ASK YOURSELF

θ: what is the joint's angle?

d: does anything slide along z?

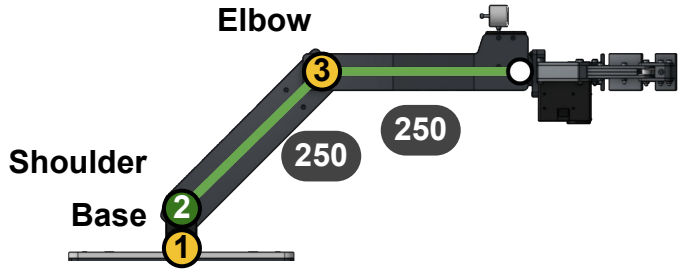
a: how long is the link along x?

α: is the next z axis tilted?

DAY 2 · THE DH TABLE

The DH Table for Our Arm

Joint	θ	d	a	α
1 · Base	θB	0	0	90°
2 · Shoulder	90° − θS	0	250	0°
3 · Elbow	−θE	0	250	0°



θ column
These change with the pose. The rest are fixed.

Shoulder
θS is measured from vertical, θ from horizontal: θ = 90° − θS.

Elbow
The elbow bends clockwise, θ counts counter-clockwise: θ = −θE.

α = 90°
Only the base twists: the shoulder axis is tilted 90° from the base axis.

DAY 2 · WORKED EXAMPLE

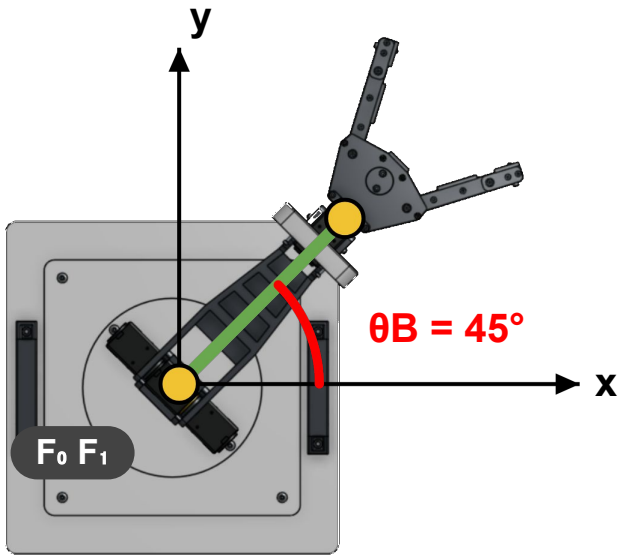
A_1: The Base

Pose: θB = 45°, θS = 45°, θE = 45°

ROW 1 $\theta = \theta B = 45° \cdot d = 0 \cdot a = 0 \cdot \alpha = 90°$

$$A_1 = \begin{bmatrix} 0.7071 & 0 & 0.7071 & 0 \\ 0.7071 & 0 & -0.7071 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

turns the frame where it sits



HOW TO READ IT

Last column (0, 0, 0): frame 1 sits at the same point as frame 0. Only the direction of its axes changes.

DAY 2 · WORKED EXAMPLE

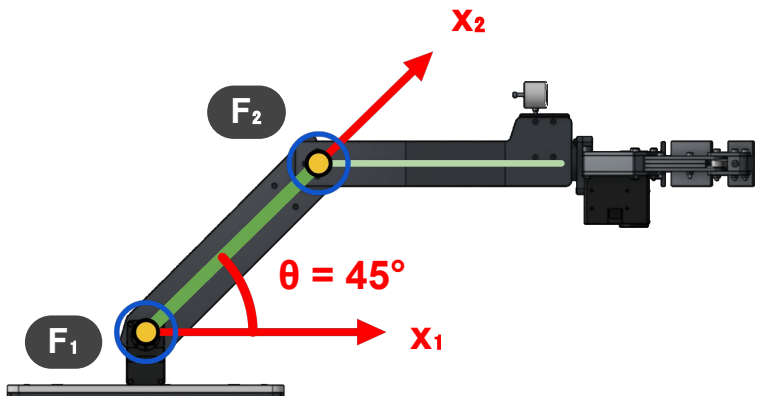
A_2: The Shoulder

Pose: θB = 45°, θS = 45°, θE = 45°

ROW 2 $\theta = 90° - \theta S = 45° \cdot d = 0 \cdot a = 250 \cdot \alpha = 0°$

$$A_2 = \begin{bmatrix} 0.7071 & -0.7071 & 0 & 176.78 \\ 0.7071 & 0.7071 & 0 & 176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

turns the frame where it sits



HOW TO READ IT

Last column (176.78, 176.78, 0): frame 2 sits 176.78 mm along x_1 and 176.78 mm along y_1, measured in frame 1.

DAY 2 · WORKED EXAMPLE

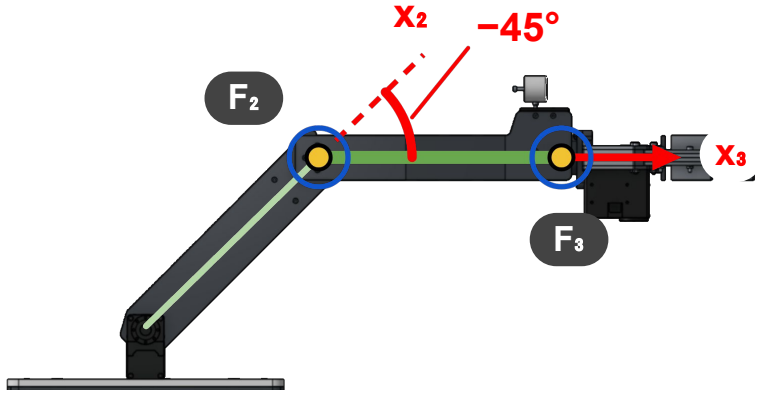
A_3: The Elbow

Pose: θB = 45°, θS = 45°, θE = 45°

ROW 3 $\theta = -\theta E = -45° \cdot d = 0 \cdot a = 250 \cdot \alpha = 0°$

$$A_3 = \begin{bmatrix} 0.7071 & 0.7071 & 0 & 176.78 \\ -0.7071 & 0.7071 & 0 & -176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

turns the frame where it sits



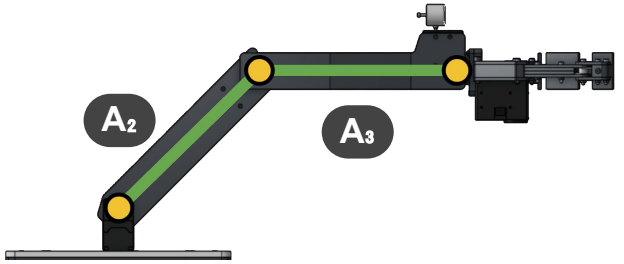
HOW TO READ IT

Last column (176.78, −176.78, 0): frame 3 sits 176.78 mm along x_2 and −176.78 mm along y_2, measured in frame 2.

DAY 2 · WORKED EXAMPLE

Step 1: Multiply $A_2 \cdot A_3$

A_2 and A_3 have parallel z axes, so together they are the flat arm of Day 1.



$$\underbrace{\begin{bmatrix} 0.7071 & -0.7071 & 0 & 176.78 \\ 0.7071 & 0.7071 & 0 & 176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_2} \times \underbrace{\begin{bmatrix} 0.7071 & 0.7071 & 0 & 176.78 \\ -0.7071 & 0.7071 & 0 & -176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_3} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 426.78 \\ 0 & 1 & 0 & 176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_2 \cdot A_3}$$

TOP-LEFT BLOCK

It is the identity matrix. The turns cancel: +45° and −45° make 0°, so frame 3 points the same way as frame 1.

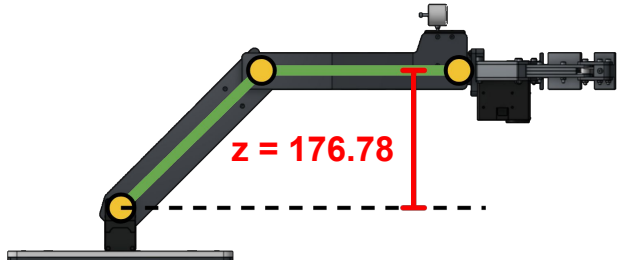
LAST COLUMN

(426.78, 176.78, 0) is Day 1's answer: the tip is 426.78 mm out and 176.78 mm up, in the plane of the arm.

DAY 2 · WORKED EXAMPLE

Step 2: Multiply $A_1 \cdot (A_2 \cdot A_3)$

Now add the base turn. The last column is the answer.



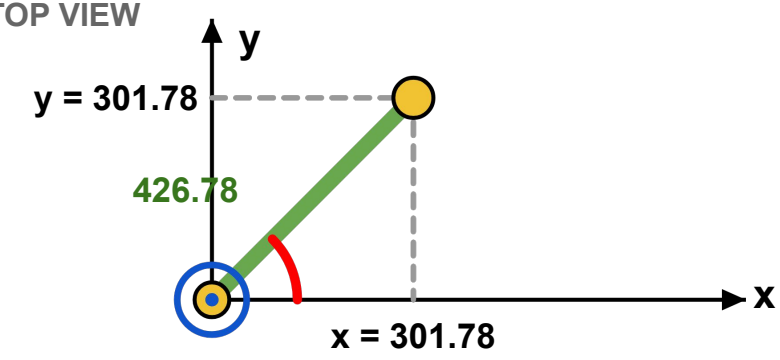
$$\underbrace{\begin{bmatrix} 0.7071 & 0 & 0.7071 & 0 \\ 0.7071 & 0 & -0.7071 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_1} \times \underbrace{\begin{bmatrix} 1 & 0 & 0 & 426.78 \\ 0 & 1 & 0 & 176.78 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_2 \cdot A_3} = \underbrace{\begin{bmatrix} 0.7071 & 0 & 0.7071 & 301.78 \\ 0.7071 & 0 & -0.7071 & 301.78 \\ 0 & 1 & 0 & 176.78 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{T = A_1 \cdot A_2 \cdot A_3}$$

THE TIP POSITION

(x, y, z) = (301.78, 301.78, 176.78) mm

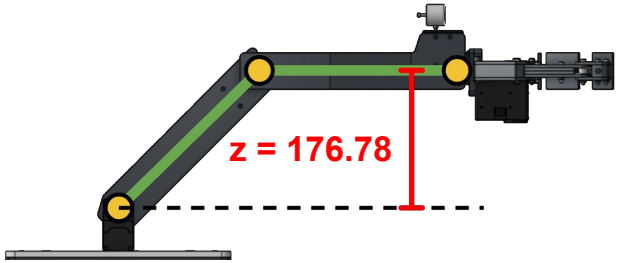
Day 1 gave a reach of 426.78 mm. The base turn of 45° splits it: 426.78 × cos 45° = 301.78 for x, and 426.78 × sin 45° = 301.78 for y. The height z = 176.78 does not change.

TOP VIEW



DAY 2 · CHECK

Same Answer as Lesson 3



LESSON 3 · TRIGONOMETRY

Triangles give a reach R and a height h.

X = 301.78

Y = 176.78 up

Z = 301.78

=

LESSON 4 · DH MATRICES

The last column of $T = A_1 \cdot A_2 \cdot A_3$.

x = 301.78

y = 301.78

z = 176.78 up

SAME NUMBERS, DIFFERENT LETTERS

Lesson 3 put up on Y. Lesson 4 puts up on z. So Lesson 3's Y is z here, and its Z is y.

WHY THEY MATCH

Both methods describe the same arm in the same pose, so they land on the same point.

DAY 2 · TRY IT YOURSELF

Try It Yourself #4

Find the tip position (x, y, z).

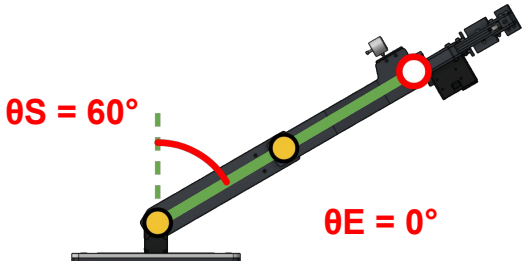
$L_1 = L_2 = 250$ mm

$\theta B = 30°$

$\theta S = 60°$

$\theta E = 0°$

(x, y, z) = (?, ?, ?)



Joint	θ	d	a	α
1 · Base	?	0	0	**90°**
2 · Shoulder	?	0	250	0°
3 · Elbow	?	0	250	0°

SOLUTION

θ column: 30°, 90° − 60° = 30°, 0°

θ = 90° − θS for the shoulder and θ = −θE for the elbow.

TIP POSITION (x, y, z) IN MM

(375.00, 216.51, 250.00)

The same arm shape as Try It Yourself #3, with the base turned 30°.

DAY 2 · TRY IT YOURSELF

Try It Yourself #5

Find the tip position (x, y, z).

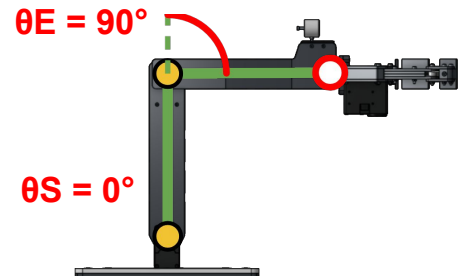
$L_1 = L_2 = 250$ mm

$\theta B = 120°$

$\theta S = 0°$

$\theta E = 90°$

(x, y, z) = (?, ?, ?)



Joint	θ	d	a	α
1 · Base	?	0	0	**90°**
2 · Shoulder	?	0	250	0°
3 · Elbow	?	0	250	0°

SOLUTION

θ column: 120°, 90° − 0° = 90°, −90°

θB = 120° turns the arm past the y axis, so x is negative.

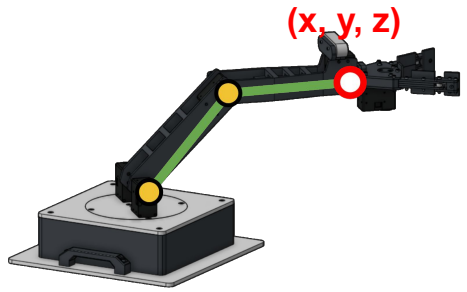
TIP POSITION (x, y, z) IN MM

(−125.00, 216.51, 250.00)

z = 250: the upper arm stands straight up and the forearm lies flat.

DAY 2 · CHECK WITH CODE

Check It with Code



PYTHON

```python
from dh_helper import dh_matrix, position
thetaB, thetaS, thetaE = 45, 45, 45
A1 = dh_matrix(thetaB, 0, 0, 90)
A2 = dh_matrix(90 - thetaS, 0, 250, 0)
A3 = dh_matrix(-thetaE, 0, 250, 0)
T = A1 @ A2 @ A3
print(position(T))
```

OUTPUT

```
[301.78, 301.78, 176.78]
```

The same as the hand calculation.

CHECK YOUR ANSWERS

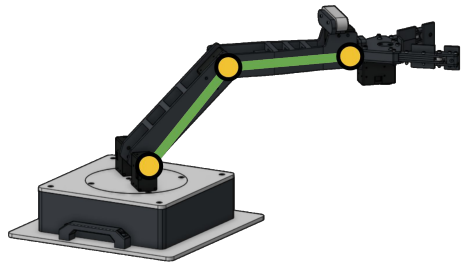
Change the three angles in line 2 to the values from Try It Yourself #4 and #5.

WHAT EACH PART DOES

dh_matrix builds one matrix from one row of the table. @ multiplies matrices. position reads the last column.

DAY 2 · COMPARISON

Trigonometry vs. DH Matrices



LESSON 3 · TRIGONOMETRY

- \+ Quick for one or two joints
- \+ Easy to picture with triangles
- − New triangles for every new joint
- − Gets messy when the arm tilts in 3D

LESSON 4 · DH MATRICES

- \+ The same recipe for every joint
- \+ A computer can do the multiplying
- \+ A standard way to describe robot arms
- − More setup: frames and a table

Both methods give the same answer. DH keeps working as the arm gets bigger.

DAY 2 · WRAP-UP

What We Learned

1. Every joint becomes one matrix, built from four numbers: θ, d, a, α.
2. Frames: z is the joint axis, x is the link.
3. Multiply the matrices in joint order: $T = A_1 \cdot A_2 \cdot A_3$.
4. The last column of T is the tip position (x, y, z).

NEXT TIME · LESSON 5

Code Implementation of FK & Other Methods

We turn forward kinematics into code, then take a short look at screw theory, another way to describe the same arm.

